Supervised ML → Ensists.

→ X

x_1 x_2 x_3 x_4

Y

Target - | label.

AdaBoost

↳ Ensemble

Bagging Boosting -

BOOTSTRAPPIN AGG.

LEARN FROM RESIDUALS!!

ENSEMBLE ⇒ MOE

DS ALGO & COMPETITIVE

DEV → PROGRAM

HACKATHONS LOOK | DEV

Check list.

① Stacking / Voting class | XG GBM Naive Bayes | NOT TODAY.

② Fastapi ✓

TOMMORROW = ACTUAL | DATA → Link → Summary

TODAY AGENDA

LEARNING | K-MEANS

(1) UNSUPERVISED | CLUSTERING.

(2) FLASK | K-MEANS | PCA

(1)
(3) GCP | AWS (2) IF TIME

DB SCANS
HIERARCHICAL SCANS

Retail Customer Segmentation.

Shopping Mall.

- Spending Score.
- Age
- Annual INCOME

Personalized Marketing Information.

	Age	Income (₹)	Score
PersonA	18	5K	5
PersonB	29	200k	70
PersonC	50	600k	50

Low Income Low Spend.

Mid Incom Mid spend

High Income Low Spending. (HOME ACCESSORIES)

How Do I Annotate this data?

(1) Manually sit & Annotate.

(2) I use Grouping algorithm finding.. → to group customers into Buckets

Goal: I have no pre-Existing labels. & I am obliged to create Groupings / Patterns from my data automatically. - Natural groups of customers

THE UNSUPERVISED LEARNING → NATURAL PATTERN

Customers
-> Budget Customers
-> Mid Spend "
-> Premium "

} k-Means clustering

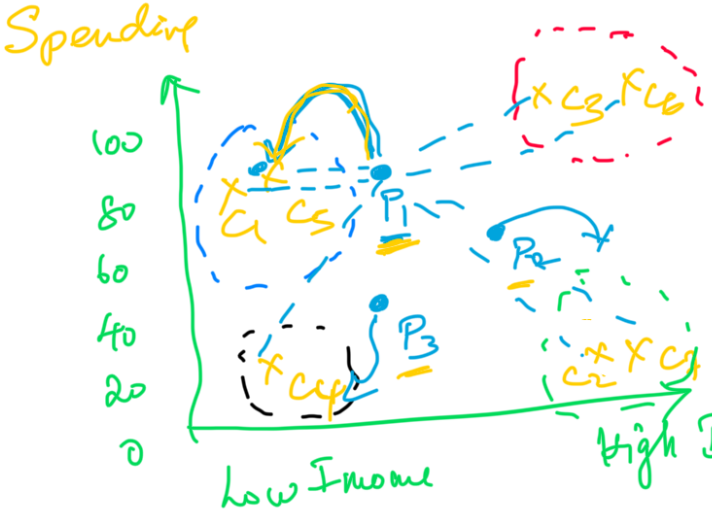
Age / Income / Spending

Customer	Age	Income	Spen.
C_1	22	25	80
C_2	45	90	15
C_3	31	85	88
C_4	48	30	20

(1) How many groups do we want to find?

No. of clusters | $k = 5$

$x_i = (\text{Income}, \text{Spending})$



Idea: k-means attempts to find these groups mathematically.

(2) Choose k value & groups

(1) Choose a [illegible]

$K = 3$

(2) Initialize K points (Initialized Randomly)

(3) Between K points & other data in the board distance.

Eucledian: $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

(4) Update / Move each point to the centroid of the closest point to it.

Centroid =

Income $\left(\frac{20+25+30}{3}\right) = 25$

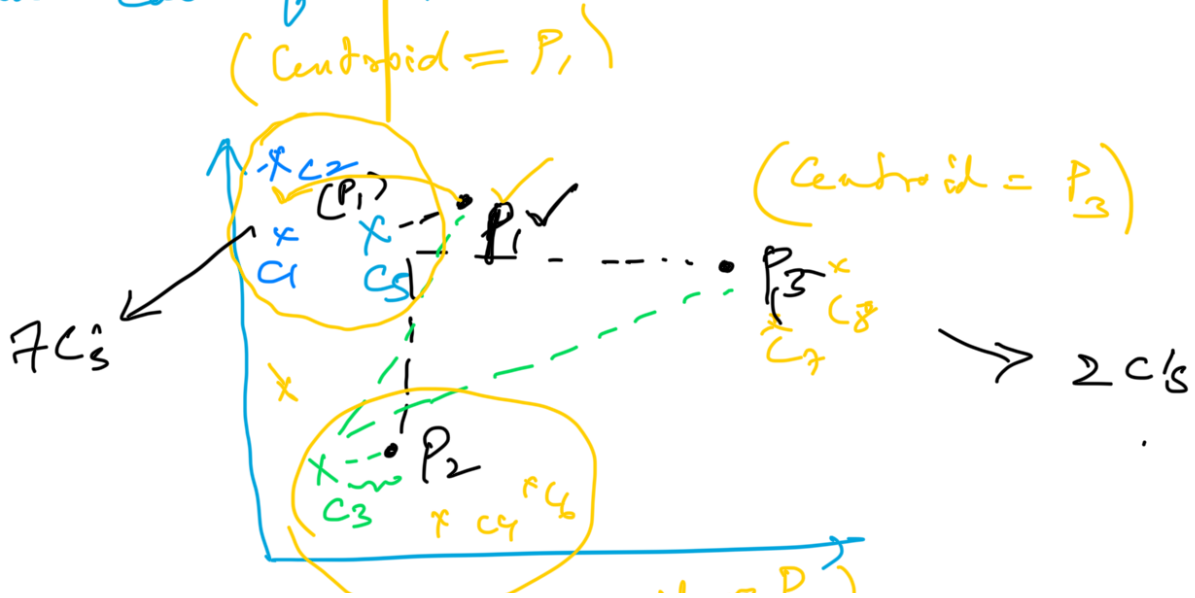
Spending $\left(\frac{70+80+75}{3}\right) = 75$

Centroid = (25, 75)

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(1) Initialization :- We begin by randomly selecting K points //

(2) Between each of K points & data ⇒ distance.

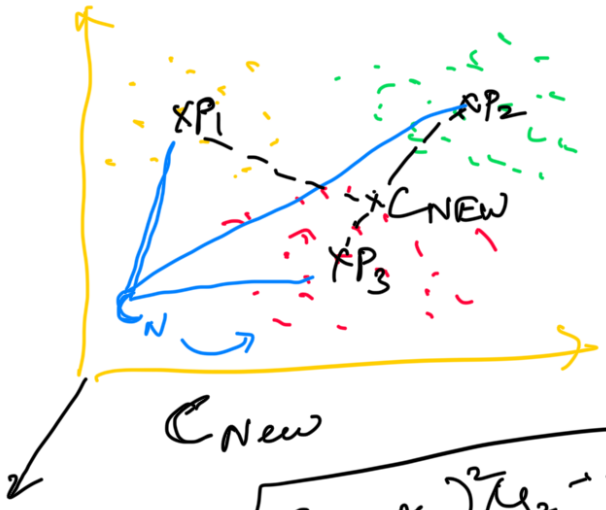


(Centroid = [2])

③ Update Step: After assigning, we recalculate the each cluster by averaging the points.

$$\text{Centroid} = \frac{(C_1 + C_2 + \dots + C_N)}{N} = P_1$$

④ Repeat: This process is repeated until the centroids no more change / maximum number of iteration is reached.



Hierarchical Clustering

If categorical is present

$$\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$$

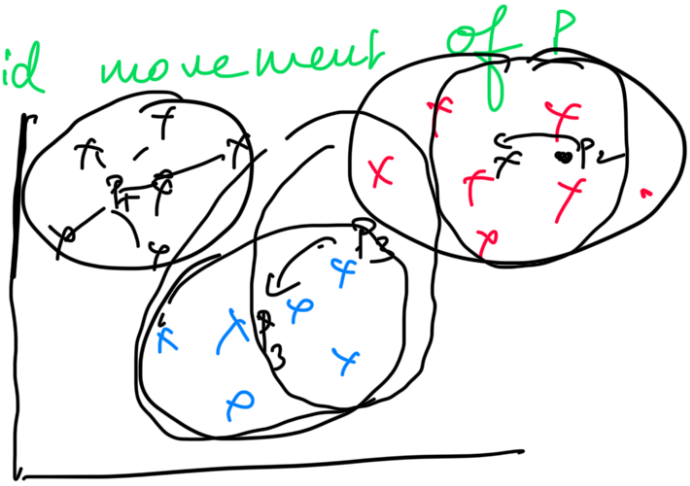
feature that describes the ⟹ business logic of the operation well.

Iteration 1 ⇒ Random P





Iteration 2 => Centroid movement of P



K=3 K=4 K=5 K=7 } Trial & Error

K = (3 to 12)

K = 17

3/4/5 } ITERATION

I find Clear groups

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(1) Practical Implementation

(2) create a flask app.

(3) deploy this in GCP.

(4) deploy => Remote Server :